

王睿 (Rita Wang)

2003-04 & +86 186 2243 2145 & ritawang97@outlook.com

教育经历

- 哈尔滨工业大学 - 计算学部 网络空间安全2026.08-2030.07
- 香港大学 - 电气与计算机工程学院 电气电子工程2025.08-2026.07
- 主修课程 模式识别与机器学习、数字图像处理与计算机视觉、数字信号处理 (A-)、生物信号与系统 (B+)
- 天津大学 - 电气自动化与信息工程学院 自动化2021.08-2025.07
- 主修课程 计算机控制、优化技术基础、模式识别基础、图像处理技术、人工智能概论、电路基础、51 单片机

工作经历

- 中关村实验室 - 岳银亮课题组 实习生2026.01-2026.05
- 事件链建模与数据构建 面向微博、B 站、抖音、小红书等多平台，构建舆情文本数据集，并基于时间顺序与语义关联建立连续事件链结构，刻画舆情演化过程；
 - 舆情演化分析 在事件链框架下分析舆情主题变化与情绪演化特征，观察不同阶段的认知变化趋势；
 - 舆情传播特征研究 结合用户互动行为与传播结构，对事件传播过程中的群体认知变化进行探索性分析。
- 香港大学电气与计算机工程学院 - Cheng Chen 课题组 研究助理2025.12-2026.05
- 多模态建模设计 构建融合影像、病理、基因及人口统计信息的生存预测模型，面向真实临床中模态不完整的数据场景进行建模；
 - 缺失模态鲁棒学习 设计基于 mean vector 融合的多模态学习框架，引入模态 dropout 与特征重建约束，有效利用不完整模态数据进行训练；
 - 实验与性能验证 在大规模真实医疗数据集上开展系统实验与消融分析，验证模型在缺失模态条件下的稳定性与预测性能提升。
- 中国人民解放军总医院 - 实习生2023.06-2023.08 / 2025.09-2026.01
- 医学数据分析与可视化 基于全球疾病负担 (GBD) 数据库计算年度变化百分比 (EAPC) 和年龄标准化率 (ASR)，进行相关性分析，并绘制散点图、密度图、疾病构成比图和趋势地图等可视化图表；
 - 心电信号与容量状态评估研究 结合 ECG 信号、BNP / NT-proBNP 及临床信息进行特征工程，基于 XGBoost 等机器学习模型开展心力衰竭容量状态预测与风险评估研究；
 - CKMM 共病演化研究 基于多中心临床数据构建共病网络，分析心血管-肾脏-代谢疾病 (CKMM) 的共病模式与演化关系，并开展相关预测模型研究 (JCR Q1 第一作者论文, DOI: 10.1016/j.maturitas.2026.108950)；
 - 口腔影像智能诊断研究 基于口内正面照片与 CBCT 影像构建数据集，利用 ResNet 等深度学习模型开展上颌横向发育不足 (MTD) 自动识别研究 (共同第一作者论文)。
- 清华大学精仪系 - 孔令杰课题组 研究助理2024.01-2024.03
- 深度学习在肝癌预后标志物识别中的应用 研究并成功复现 PathFinder 算法，用于肝癌预后标志物的识别；
 - 视觉皮层信号解码研究 复现 CEBRA 算法，研究小鼠观看视频时的视觉皮层信号解码；
 - 脑数据处理与预测 处理小鼠观看复杂视频时的脑部数据，利用 DINO 算法压缩视频为图片，并预测后续视频帧。
- ## 科研项目
- sdLDL-C 与心肾代谢多病共存纵向演化研究 (JCR Q1 第一作者)2025.09-2026.04

- 纵向多状态建模设计 基于 CHARLS 前瞻性队列数据，构建 multi-state Markov 模型，刻画健康 → 单病 → 多病共存 (CKMM) 的疾病演化轨迹；
 - 风险关联与阶段转移分析 结合 Cox 回归与转移强度估计，系统评估 sdLDL-C 与 LDL-C 在不同疾病阶段中的风险效应与转移概率差异；
 - 残余风险识别与机制评估 通过分层与敏感性分析验证 sdLDL-C 在 LDL-C 控制下的残余风险预测价值，识别其在早期转移与多病进展中的阶段特异性作用；
 - 模型验证与结果解释 完成多重插补、非线性关系检验与消融分析，确保模型稳健性并形成机制层面的风险解释框架。

基于深度学习的上颌横向发育不足智能诊断研究 (JCR Q1 共同第一作者)2025.09-2026.05

- 研究设计与数据构建 构建包含 826 例口内正面照片 +CBCT 的正畸影像数据集，以 CBCT 衍生的 UPA/YTA 指标作为金标准完成样本标注，并建立标准化医学影像数据处理流程；
- Rui Wang
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- **模型开发与对比评估** 基于 PyTorch 实现并系统评估 DenseNet、ResNet、EfficientNet、MobileNet 等多种深度学习模型，比较不同网络结构在医学影像分类任务中的诊断性能；
- **统计验证与泛化评估** 采用 5-fold 交叉验证评估模型稳定性，并通过 Bootstrap 计算 95% 置信区间；利用 DeLong 检验比较不同模型的 AUC 差异，并引入香港大学 FDTooth 外部数据集 ($n \approx 200$) 进行独立验证以评估模型泛化能力；
- **诊断性能与临床价值** 在无需 CBCT 的情况下，仅基于口内照片实现上颌横向发育不足筛查，最优模型在 YTA 标注体系下准确率达 99%+，显示出在降低辐射依赖、减少诊断成本及提升正畸初筛效率方面的潜在临床应用价值。

VideoTutor: 基于 RAG 与大语言模型辅助视频生成的自适应知识机器人增强学习系统 2025.08-2026.05

- **系统架构设计** 面向真实教学场景，设计以教学 PDF 为核心的智能教学系统架构，整合文档解析、语义检索、智能问答与多模态内容生成模块；
- **智能问答与可靠性设计** 基于 RAG 框架实现页级语义检索与可溯源问答机制，降低大模型幻觉问题，提升问答结果的准确性与可解释性；
- **多模态内容自动生成** 实现从教学 PDF 到讲解脚本、语音、教学视频及多语言字幕的端到端自动生成流程，显著提升教学内容生产效率；
- **人机协同与系统实现** 引入教师-学生双角色审核机制，支持人工修订与质量控制，基于 Python 完成系统实现与验证。

四轮驱动洗船机转速控制系统设计 - 优秀本科毕业设计 2025.01-2025.06

- **系统架构设计** 基于 eCore640S-I 工业控制器，完成整体方案与关键硬件选型；
- **控制策略研发** 提出融合滑模控制、交叉耦合控制与线性扩张状态观测器的复合同步控制策略，解决液压系统非线性及多轮协同问题，提升转速控制精度与鲁棒性；
- **建模与仿真** 利用 MATLAB/Simulink 进行系统建模仿真分析，验证所提控制方法在扰动环境下的稳态与多轮同步性能；
- **控制程序实现** 在 CoDeSys 软件环境编写控制器程序，基于状态机构建运行逻辑，实现设备启停、前进后退及速度调度控制，构建完整的闭环控制流程。
- **知识产权成果** 中国发明专利 1 项（导师为第一发明人，本人为第二发明人）。

基于人脑视觉感知的自动驾驶系统 - 核心成员 2023.03-2024.04

- **硬件与机械布局** 使用 Altium Designer 绘制系统框图并设计 PCB，规划结构，完成打孔、编码器安装、电路板布置及摄像头组装；
- **软件开发** 编写硬件驱动、速度控制及摄像头图像处理程序，实现轨迹识别与自主控制；
- **会议论文** 第一作者撰写长摘要《带有尖峰树突混合机制驱动的多尺度学习》，Paper ID: CCHI2023-102443。

竞赛经历

第二届大学生电气与电子工程创新大赛华北赛区二等奖 2023.03-2023.09

面向典型季节变化特性与源荷不确定性，构建冷-热-电-气综合能源系统优化调度模型。

校园经历

天津大学北洋军乐团 - 副团长 & 声部长 2021.08-2025.07

演出与组织协调 参与多场校内外演出，如天津大剧院和天津仁爱学院等，获北京国际音乐节“示范乐团”称号及天津市艺术展演一等奖，并全程组织策划十余场演出。

其它技能

语言 英语 (TOEFL 86)

计算机 Python、R 语言、MATLAB，熟练运用 Office 等常用的办公软件。

其它 中国音乐学院架子鼓 10 级、小鼓 7 级，辩论。

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Education

Ph.D. in Cyberspace Security, Harbin Institute of Technology 2026.08-2030.07

M.Sc. in Electrical and Electronic Engineering, The University of Hong Kong 2025.08-2026.07

Main Courses: Pattern Recognition and Machine Learning (A), Digital Image Processing and Computer Vision (A-), Digital Signal Processing (A-), Biomedical Signals and Systems (B+).

B.Eng. in Automation, Tianjin University 2021.08-2025.07

Main Courses: Introduction of Artificial Intelligence (95), Basis of Pattern Recognition (98), Image Processing Technologies (97), Basis of Optimization Technology (95), Autocontrol Theory (90).

Publication & Patent

1. **Rui Wang**, Jincheng Zhao, Yi Tang, Qingtao Zhang, Dayang Xie, Wei Dong*, Xiaoyan Li*. Association of small dense low-density lipoprotein cholesterol with longitudinal trajectories of cardio-vascular-kidney-metabolic comorbidity: A prospective cohort study based on multi-state Markov models. *Maturitas*, 2026. DOI: 10.1016/j.maturitas.2026.108950.
2. Jianing Li¹, **Rui Wang**¹, Zhou Yi, Jianhui Ni, Zhigang Zuo*, Yue Wang*. Comparing Deep Learning Models for Identifying Maxillary Transverse Deficiency from Intraoral Photographs. *International Dental Journal*. 2026;76(4):109689. DOI: 10.1016/j.identj.2026.109689.
3. **Rui Wang**, Xinyi Tang, Shuangming Yang*. Peak-driven Multi-scale Learning with a Mixing Mechanism with Spiky Dendrites. *The 5th China Symposium on Cognitive Computing and Hybrid intelligence*. Paper ID: CCHI2023-102443. 2023.09.30.
4. Donghui Li, **Rui Wang**. A Multi-Wheel Synchronous Control Method and System for Ship-Cleaning Machines Based on SMC-CCC-LESO. Chinese Patent, under substantive examination.

Internships

Intern, Prof. Yinliang Yue's Group, Zhongguancun Laboratory 2026.01-05

- **Event-chain modeling and data construction:** built cross-platform brand-related event-chain datasets from Weibo, Bilibili, Douyin, and Xiaohongshu to characterize temporal evolution of public brand cognition under consecutive events.
- **Group cognition and polarization analysis:** analyzed cognitive divergence paths of different user groups by integrating user behavior features and interaction networks; quantified stance shifts, emotional accumulation, and polarization dynamics.
- **Cognitive security mechanism modeling:** investigated polarization mechanisms from event sequencing, group heterogeneity, and cognitive-object migration perspectives to support brand-related cognitive risk identification and early warning.

Research Assistant, Prof. Cheng Chen's Group, The University of Hong Kong 2025.12-2026.05

- **Multimodal modeling for survival prediction:** developed survival prediction models integrating radiology, pathology, genomic, and demographic data, targeting real-world clinical scenarios with incomplete modalities.
- **Robust learning under missing modalities:** designed a mean-vector-based multimodal fusion framework with modality dropout and feature reconstruction constraints to effectively leverage incomplete multimodal data.
- **Experimental validation and analysis:** conducted large-scale experiments and ablation studies on real medical datasets to evaluate model robustness and performance gains under missing-modality conditions.

Intern, Chinese People's Liberation Army General Hospital 2023.06-08, 2025.09-2026.01

- **Medical data analysis and visualization:** Calculated Annual Percentage Change (EAPC) and Age-Standardized Rate (ASR) using the Global Burden of Disease (GBD) database; conducted correlation analyses and generated visualizations including scatter plots, density plots, disease composition charts, and global trend maps.

- **ECG-based volume status assessment:** Performed feature engineering by integrating ECG signals, BNP/NT-proBNP biomarkers, and clinical variables; developed machine learning models (e.g., XGBoost) for predicting volume status and risk assessment in heart failure patients.
- **CKMM comorbidity evolution analysis:** Constructed comorbidity networks using multi-center clinical data to investigate the evolution patterns of Cardiovascular-Kidney-Metabolic (CKMM) diseases and developed predictive models for disease progression (first-author paper, JCR Q1).
- **Oral image-based intelligent diagnosis:** Built a dataset of frontal intraoral photographs paired with CBCT images and applied deep learning models (e.g., ResNet) to automatically identify Maxillary Transverse Deficiency (MTD) (co-first-author paper).

Research Assistant, Prof. Lingjie Kong's group, Tsinghua University

2024.01-03

- **Application of deep learning in liver cancer prognostic biomarker identification:** studied and successfully reproduced the PathFinder algorithm for biomarker recognition.
- **Visual cortex signal decoding:** reproduced the CEBRA algorithm to analyze decoding in mice during video viewing.
- **Brain data processing & prediction:** processed neural data from mice exposed to complex videos; used the DINO algorithm to compress videos into image representations and predict subsequent frames.

Research Projects

Longitudinal Trajectory Modeling of Cardiovascular-Kidney-Metabolic Multimorbidity Associated with sdLDL-C (First Author, JCR Q1)

2025.09-2026.04

- **Longitudinal Multi-state Modeling Design:** Constructed a multi-state Markov model based on the prospective CHARLS cohort to characterize disease progression trajectories from healthy state → single disease → cardiovascular-kidney-metabolic multimorbidity (CKMM).
- **Risk Association and Stage-specific Transition Analysis:** Integrated Cox proportional hazards models with transition intensity estimation to systematically evaluate the stage-dependent effects of sdLDL-C and LDL-C on disease progression probabilities.
- **Residual Risk Identification and Mechanism Assessment:** Conducted stratified and sensitivity analyses to validate the residual predictive value of sdLDL-C beyond LDL-C control, identifying its stage-specific role in early transition and multimorbidity progression.
- **Model Validation and Robustness Evaluation:** Performed multiple imputation, nonlinear relationship testing, and model robustness analyses to ensure statistical reliability and mechanistic interpretability of findings.

Deep Learning-Based Intelligent Diagnosis of Maxillary Transverse Deficiency (Co-First Author, JCR Q1)

2025.09-2026.05

- **Study Design and Dataset Construction:** Built an orthodontic imaging dataset consisting of 826 intraoral frontal photographs paired with CBCT scans. Used CBCT-derived UPA/YTA measurements as reference standards for sample labeling and established a standardized medical image preprocessing pipeline.
- **Model Development and Comparative Evaluation:** Implemented and systematically evaluated multiple deep learning architectures using PyTorch, including DenseNet, ResNet, EfficientNet, and MobileNet, to compare their performance in medical image classification tasks.
- **Statistical Validation and Generalization Assessment:** Employed 5-fold cross-validation to evaluate model robustness and calculated 95% confidence intervals using bootstrap resampling. Conducted DeLong tests to compare AUC differences between models, and performed independent validation on the HKU FD-Tooth external dataset ($n \approx 200$) to assess model generalizability.
- **Diagnostic Performance and Clinical Significance:** Achieved $>99\%$ classification accuracy under the YTA labeling scheme using only intraoral photographs without CBCT imaging, demonstrating strong potential for reducing radiation exposure, lowering diagnostic costs, and improving efficiency in orthodontic screening.

VideoTutor: An Adaptive Knowledge Bot for Enhanced Learning via RAG and LLM-Assisted Video Generation

2025.08-2026.05

- **System architecture design:** designed an intelligent teaching system centered on instructional PDFs, integrating document parsing, semantic retrieval, RAG-based question answering, and multimodal content generation for real-world educational scenarios.
- **Reliable QA with RAG:** implemented page-level semantic retrieval and source-grounded answer generation under the RAG framework to reduce LLM hallucinations and improve answer accuracy and interpretability.

- **End-to-end multimodal content generation:** built an automated pipeline from teaching PDFs to presentation scripts, TTS audio, instructional videos, and multilingual subtitles, significantly improving content production efficiency.
- **Human-in-the-loop system implementation:** introduced a teacher–student dual-role review mechanism for manual revision and quality control; implemented and validated the full system in Python.

Design of Speed Control System for Four-Wheel Drive Ship Washer - Excellent Graduation Project

2025.01-06

- **System Architecture:** Designed overall scheme and key hardware based on eCore640S-I industrial controller.
- **Control Strategy:** Developed a composite synchronous control method integrating sliding mode control, cross-coupling, and LESO to address hydraulic nonlinearity and multi-wheel coordination, improving speed accuracy and robustness.
- **Modeling & Simulation:** Built system models in MATLAB/Simulink to verify steady-state and synchronization performance under disturbances.
- **Control Implementation:** Programmed in CoDeSys using state machines for logic control, enabling start/stop, forward/reverse, and speed scheduling in a closed-loop system.
- **Intellectual Property:** Co-inventor (2nd inventor) of a Chinese invention patent related to the proposed multi-wheel synchronous control method (first inventor: supervisor); patent has passed preliminary examination and is currently under substantive review.

Brain-Inspired Vision-Based Autonomous Driving System - Core Member

2023.03-2024.04

- **Hardware & Layout:** Designed system block diagram and PCB in Altium Designer; completed drilling, encoder installation, circuit board layout, and camera assembly.
- **Software Development:** Programmed hardware drivers, speed control, and camera-based image processing for trajectory recognition and autonomous control.
- **Conference Paper:** First author of extended abstract “Multi-Scale Learning Driven by Spiking-Dendritic Hybrid Mechanism”, Paper ID: CCHI2023-102443.

Competitions

the 2nd prize in the North China Area, the 2nd University Electrical and Electronic Engineering Innovation Competition

2023.03-09

Built integrated energy optimal scheduling model for cold-heat-electric-gas systems for typical seasonal variability and source-load uncertainty.

Activities & Volunteering Experiences

Deputy Leader & Section Leader of Peiyang Wind Orchestra

2021.08-2025.07

Performance and Event Coordination: Participated in multiple on- and off-campus performances, including at Tianjin Grand Theatre and Tianjin Ren'ai College; awarded “Model Orchestra” at the Beijing International Music Festival and First Prize at the Tianjin Art Exhibition; organized and coordinated over ten performances.

Other Information

Skills: R Studio, Python, and MatLab

Hobbies: percussion music (the Chinese Conservatory of Music 10th level in drum set), and debate.